GOSIM SHENZHEN 2026

Physical Interfaces for Agentic Development

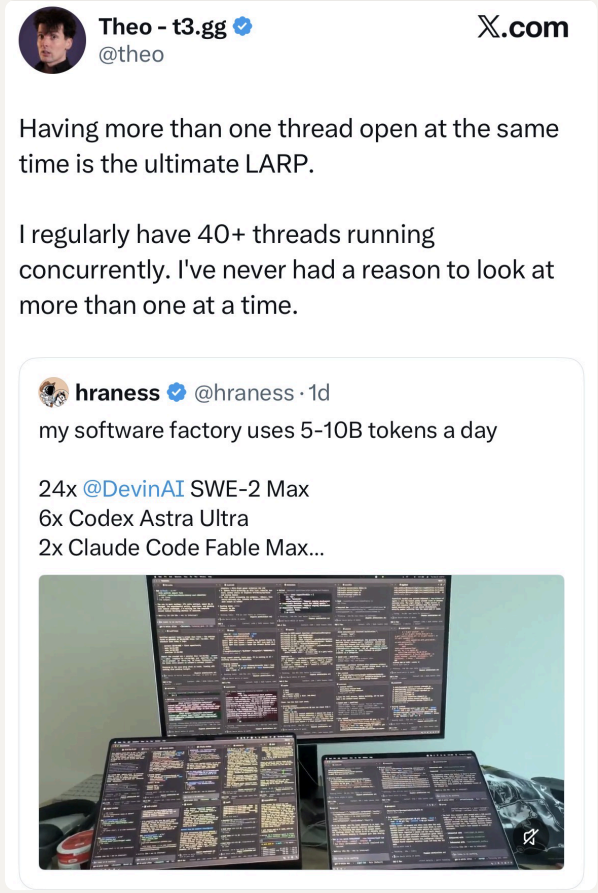
Peripheral awareness for the developer who runs more than one agent

Michael Yuan, PhD github.com/juntao GOSIM, ShenZhen 2026

MANY AGENTS, ONE PAIR OF EYES

Developers run dozens of agents. Nobody can watch them all.

> *"Having more than one thread open at the same time is the ultimate LARP."*
>
> Theo, t3.gg



47 s

Average screen attention span for knowledge workers, down from 2.5 minutes in 2004

Gloria Mark, UC Irvine

23 min

Average time to return to an interrupted task

Mark, Gudith, Klocke

3–5

Agents an engineer can supervise before quality drops

Teams at OpenAI and elsewhere

SECTION 01

The supervision problem

THESIS

The bottleneck in agentic development is now the human's attention budget.

Three acts inside "checking on an agent"

01

Notice

State changed

```
apply patch to
src/auth/session.rs?
```

Read

What does it want

03

Act

Approve, reject, redirect

Only the middle one needs a screen.

The approval prompt is the worst case



14:52

BLOCKED

An approval prompt is both an interrupt and a blocking decision. The agent waits until you find it.

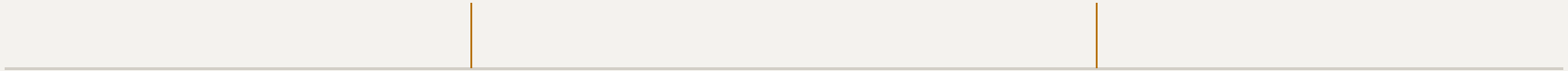
Poll versus push

POLL — YOU GO TO THE STATE



Costs a switch every time, whether or not anything changed.

PUSH — THE STATE COMES TO YOU



Costs nothing until something changes.

SECTION 02

Field guide to 2026 agentic developer's hardware

The year the agent got its own periphery device

Codex Micro

maker	OpenAI with Work Louder
price	$230
launch	July 15, 2026
input	6 Agent Keys, 6 Command Keys, joystick, reasoning dial
link	USB

Reportedly sold out in about twelve hours, with resales reported up to $1,850.





Claude Desktop Buddy

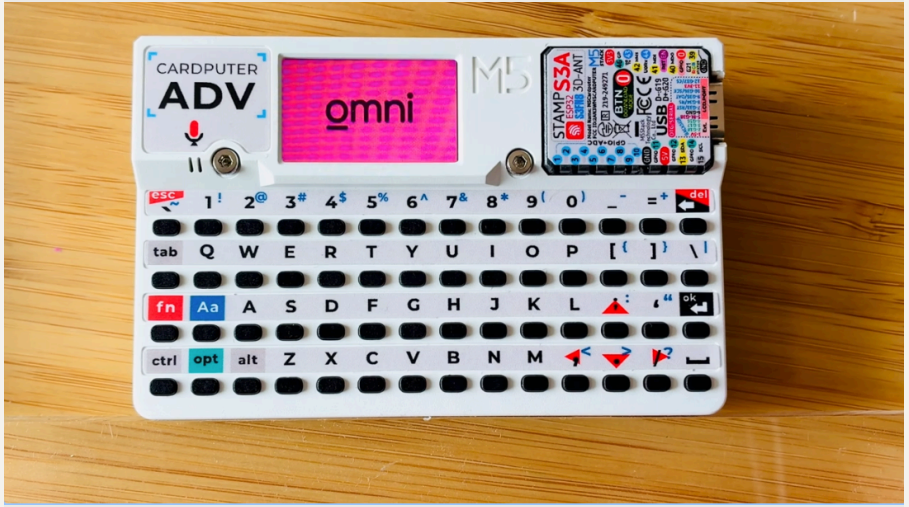
origin	Anthropic reference project
api	BLE, Claude Desktop developer mode
board	M5StickC Plus, about $20
input	two buttons, approve and deny



Seven states: sleep, idle, busy, attention, celebrate, dizzy, heart.

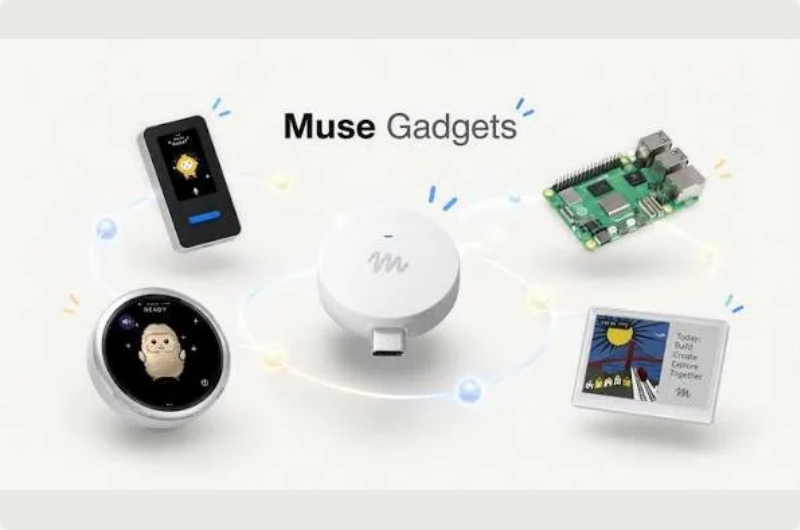
Cardputer

A full handheld computer as an agent companion. Built with UIFlow and MicroPython, with Claude Code doing the flashing and iteration.



The community ported the buddy to the StickC Plus2, StickS3, Paper, and Cardputer within weeks.

Muse Gadgets and Muse Charm



MUSE GADGETS · META

Build your own Muse device on an off-the-shelf ESP32 board or a Raspberry Pi. Open source SDKs and firmware, Apache 2.0.



MUSE CHARM · META

Meta's first AI agent device. Voice, touch and camera in a pocket charm. Over $250.

The shared pattern

- A status vocabulary of five states, rendered in color or posture.
- One physical gesture for the one decision that blocks the agent.
- Premium prices. People pay for this.
- An ESP32 underneath.

  `cheap` `Wi-Fi · BLE · USB · audio` `large community` `Rust via esp-rs` `HTTP · MQTT · BLE HID`

- Publish the protocol and the hardware design.

OSC 7501 · MITCHELL HASHIMOTO · OCTOBER 6, 2026

A terminal protocol for program status

The creator of Ghostty proposes a standard way for any program to tell the terminal what state it is in. Today, tools guess by reading the screen.

> *"I'd like us all to stop guessing what programs are doing by reading their screens or process trees."*

```
ESC ] 7501 ; state=blocked
  :kind=permission
  :app=claude-code
```

- idle
- working
- done
- blocked
- error

mitchellh.com/writing/program-status-osc7501

SECTION 03

VibeBuddy, a desk buddy for the notification problem



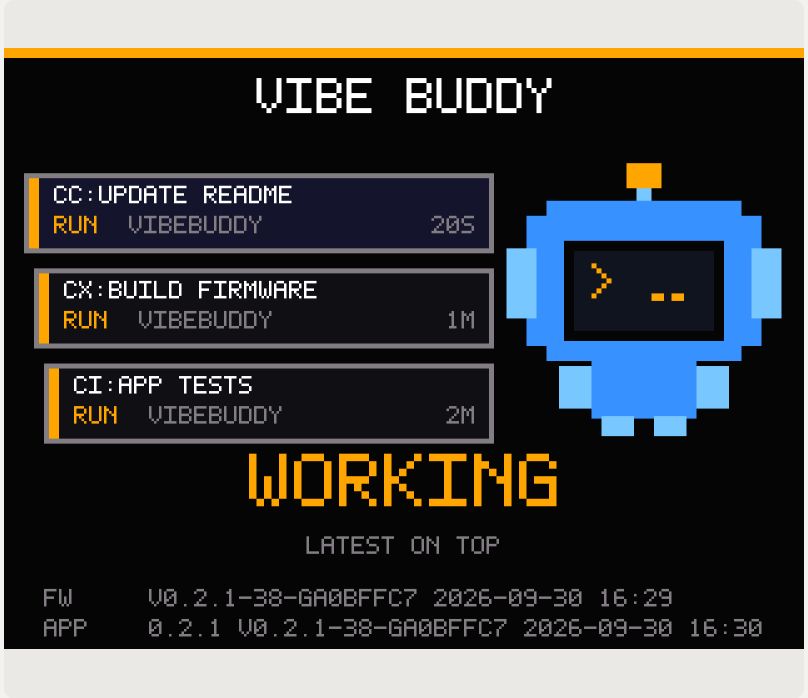


VibeBuddy

chip	ESP32-S3, 8 MB PSRAM
screen	320 × 240 color LCD
audio	speaker, spoken alerts
link	one USB-C cable
watches	Claude Code, Codex, GitHub Actions
price	$69

Your agent comes to you when it needs you in the loop.

Every agent on one screen



- `working`
  Silent. Keep coding.
- `input required`
  Said out loud, once. K2 brings the session to the front.
- `done`
  You hear it, and see how long it took.
- `failed`
  You find out now, not at the next tab check.

```
curl -d '{"event":"task.done",
          "title":"Hello"}' \
  127.0.0.1:7331/v1/events
```

The VibeBuddy app

One install hooks every agent on the machine.

AGENTS

Claude Code

Codex

OpenCode

GitHub Copilot CLI

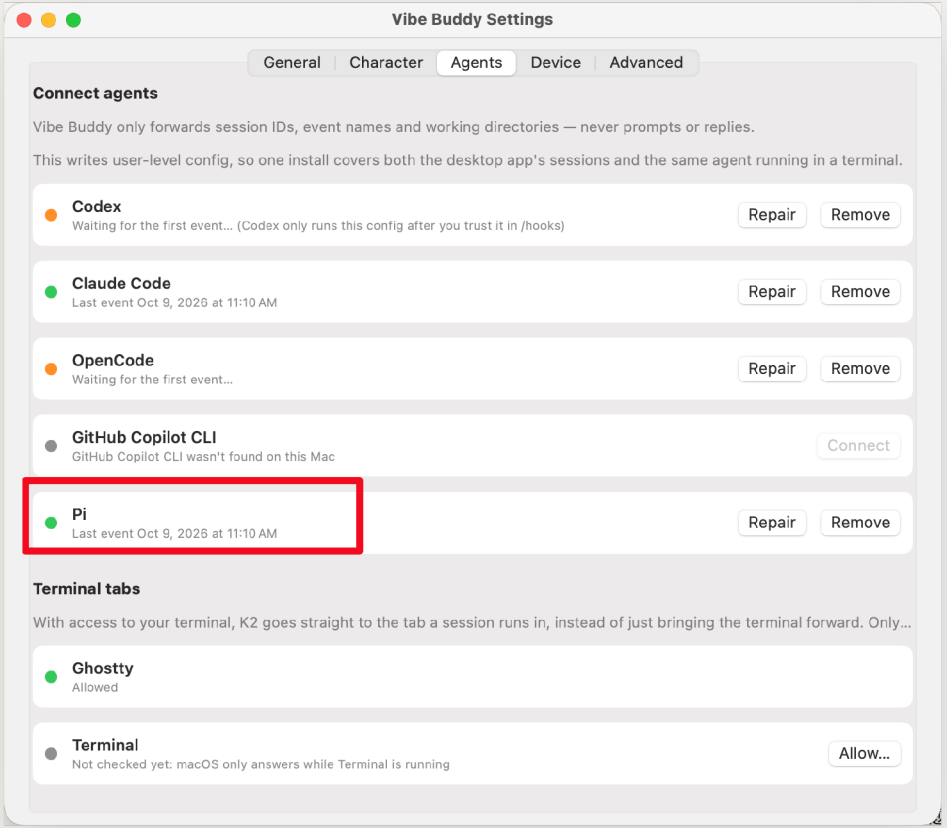
Pi

TERMINALS

Ghostty

Terminal

Forwards session IDs, event names and working directories. Never prompts or replies.



Agent status via OSC 7501 · mitchellh.com/writing/program-status-osc7501

VibeBuddy software stack

AGENTS	Claude Code · Codex · GitHub Actions · any HTTP client	–
HOOKS	Claude Code and Codex hooks · gh CLI watcher	github.com/second-state/vibebuddy
APP	macOS and Linux menu bar app · daemon · HTTP API on 127.0.0.1:7331	github.com/second-state/vibebuddy
LINK	One USB-C cable for power and events	–
FIRMWARE	Rust on ESP32-S3 · LCD UI · spoken alerts · Pomodoro	github.com/second-state/vibebuddy

GPL-3.0 code · CC BY-SA 4.0 voices

SECTION 04

VibeKeys Max, answer your agents on the go

1,000+ devices in daily use

IN PRODUCTION AT

Nansen Prisma UCLOUD优刻得 HL7 International



Hardware

CORE

ESP32-S3 with PSRAM

TFT status screen

I2S microphone

Hot-swappable mechanical keys

Rotary knob

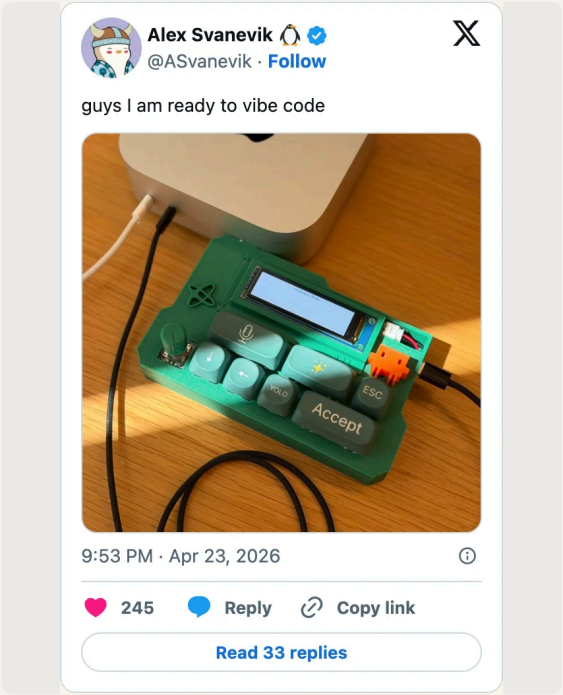
LINK & BODY

Bluetooth 5, Wi-Fi, USB-C

Battery, 15 days standby

130 × 90 × 30 mm, 150 g

$99

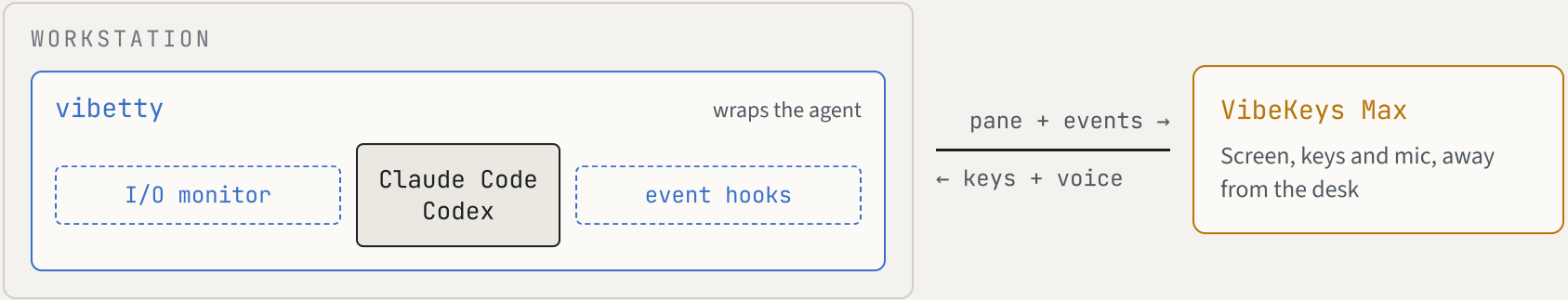


A case printed to order



Every unit is 3D printed. Pick a color at order time. The case files are open.

Full remote mode



Screen

The live agent pane and its current event

Keys

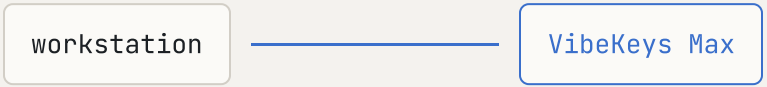
Approve, reject, interrupt, retry

Voice

Speak the next prompt into the mic key

Over Wi-Fi, on the local network or the public internet

AT HOME OR THE OFFICE



Same Wi-Fi

The device reaches vibetty on the local network.

ANYWHERE ELSE



Over Tailscale

Both ends join one tailnet. WireGuard encryption, no port forwarding.

VibeKeys Max software stack

AGENTS	Claude Code · Codex · Herdr · Omarchy	–
VIBETTY	Wraps the agent · terminal I/O monitor · event hooks	github.com/second-state/vibetty
HOST APP	CLI · Claude Code, Codex, Herdr and Omarchy plugins	github.com/second-state/vibekeys_app
NETWORK	Local Wi-Fi or Tailscale over the internet	–
FIRMWARE	Rust on ESP-IDF · keys and screen · esp-sr audio to Whisper · OTA	github.com/second-state/vibekeys_firmware

Target xtensa-esp32s3-espidf · OTA from GitHub releases

One more thing

ONE MORE THING

Omarchy, a malleable OS for the age of agents

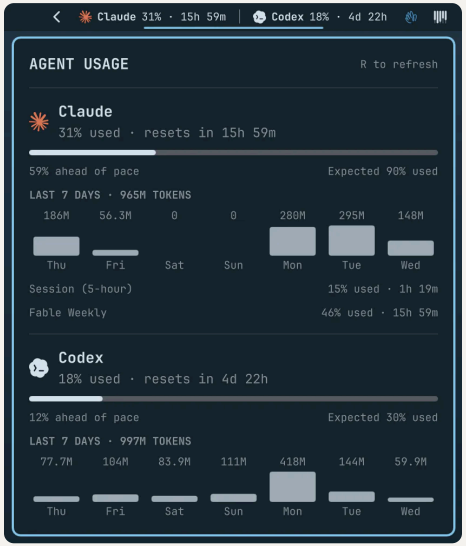
DHH's Arch-based desktop. Ships with coding agents wired into the OS.

OMARCHY EDITIONS

Special editions of VibeBuddy and VibeKeys, made for Omarchy



Seven agent sessions running at once, on one desktop



The agent in the bar

Build your own

`github.com/second-state/vibekeys_firmware`

Rust firmware

`github.com/second-state/vibekeys_app`

Host CLI and Claude Code / Codex plugins

`github.com/second-state/vibebuddy`

VibeBuddy firmware, daemon, hooks and protocol

`github.com/second-state/vibetty`

Agent wrapper: I/O monitor, event hooks, remote mirror

`github.com/anthropics/claude-desktop-buddy`

Reference buddy firmware and BLE protocol

`omarchy.org`

The malleable OS for the age of agents